# AI-Powered Image Classification System — Fast, Accurate, Comparative

Introduces a system that maps images to seven categories using hash-based and deep learning models. Team: Emir Yumrukkaya, Bunyamin Saka, Mustafa Deniz Vardal.



# Supported Categories in Image Database

Seven categories, 99 reference images total to support robust classification

| Category | Reference Images |
| --- | --- |
| Cat | 12 |
| Dog | 19 |
| Orange | 9 |
| Watermelon | 17 |
| Computer | 16 |
| Rock | 12 |
| Total | 99 |

# System Architecture Overview

Modular image-classification pipeline: standardized input → dual classifiers → JSON results

### Input Image

Raw image enters pipeline; source-agnostic intake.

### Preprocessing

Resize and convert to **RGB** to standardize size and color.

### Hash-based Classifier

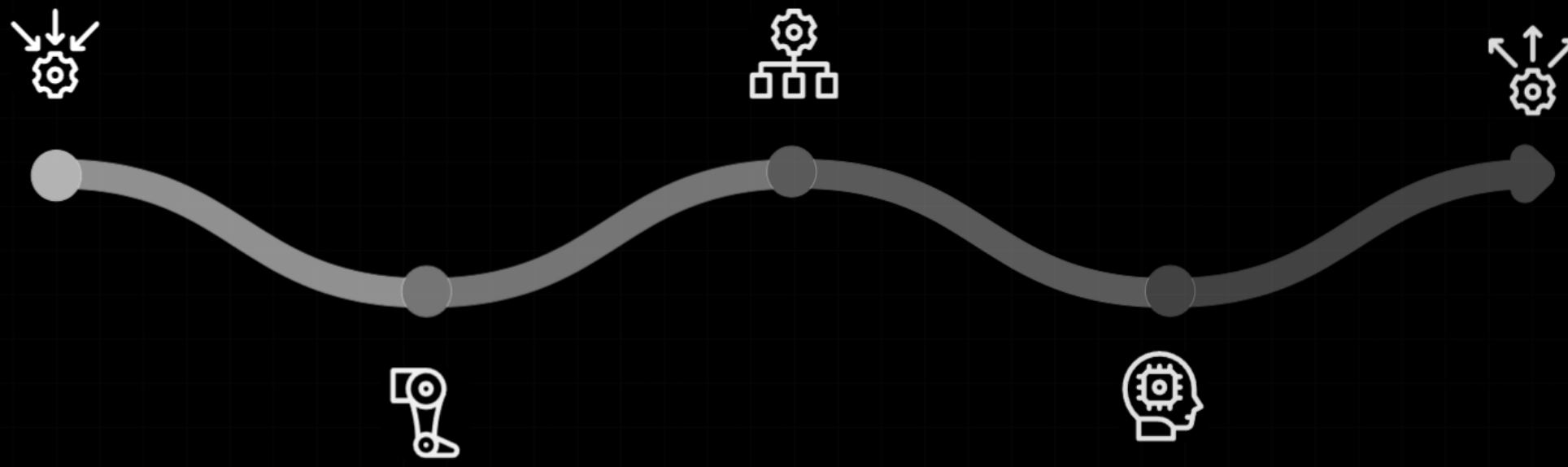
Deterministic, fast classification using hashing.

### AI-driven Classifier

Model-based prediction for nuanced recognition; returns confidence scores.

### JSON Output

Emits predicted category and confidence in **JSON** for easy integration.



# Hash-Based Image Classifiers: Fast, Structural & Color Cues

Lightweight hashing techniques for real-time or resource-constrained image classification

| Technique | Primary Purpose | Strengths | Limitations |
| --- | --- | --- | --- |
| Perceptual Hash | Capture structural similarity | Fast, robust to minor edits | Sensitive to major transforms |
| Average Hash | Emphasize overall brightness | Computationally cheap | Losesspatial detail, lower for complex categories |
| Difference Hash | Edge detection / gradients | Good for detecting structural changes | Affected by noise and heavy compression |
| Color Histogram | Capture color distribution | Preserves color info for categories | Insensitive to spatial layout |

# AI Classification Pipeline: ResNet50 with DCT Hashing

Preprocess images, extract low-frequency features, and feed robust ResNet50 model for high-accuracy classification

- Resize to 32×32
  - Downscale original image to 32×32 to standardize input and reduce compute
- Convert to grayscale
  - Remove color channels to focus on luminance-based features
- Apply DCT
  - Perform Discrete Cosine Transform to separate frequency components
- Retain low-frequency 8×8
  - Keep 8×8 low-frequency block to emphasize dominant features and reduce noise
- Generate binary hash
  - Compare coefficients to median to form a binary hash representing the image
- Feed ResNet50
  - Use ResNet50 with residual connections for robust, high-accuracy classification
- Trade-offs
  - Deep models need more compute and training data but offer higher accuracy and robustness

# Technology Stack Overview

Core tools enabling rapid prototyping and production-ready image and ML workflows

| Category | Technology / Format |
| --- | --- |
| **Programming Language** | Python 3.9+ |
| **Image Processing** | Pillow, OpenCV |
| **Perceptual Hashing** | ImageHash |
| **Deep Learning** | PyTorch, TorchVision |
| **Data Formats** | JSON, Pickle |
| **Design Goal** | Balance rapid prototyping and production readiness |

# High-Confidence Image Classification: Cat (91.6%)

Detailed category confidences for automated tagging and thresholding

| Category | Confidence (%) | Notes |
| --- | --- | --- |
| Cat | 91.6 | **High confidence** — primary label for tagging |
| Dog | 4.2 | Low likelihood |
| Rabbit | 1.1 | Very low likelihood |
| Bird | 0.9 | Very low likelihood |
| Other | 2.2 | Residual probability for rare classes |

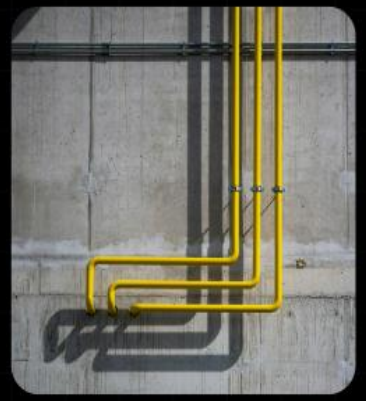
# Key Challenges & Solutions

Practical fixes for format variability, latency, and accuracy/speed trade-offs

| Challenge | Solution |
| --- | --- |
| Handling different image formats (JPG, PNG, WEBP, AVIF) | Support multiple codecs and normalize formats before classification |
| Varying image sizes and dimensions | Resize and normalize images prior to inference |
| Model loading latency | Load model once and classify multiple images in batches |
| Balancing accuracy and speed | Offer two classifiers: fast hash-based method and precise AI classifier |
| Practical deployment across diverse datasets | Combine multi-codec support, resizing, batching, and classifier options for adaptability |

# Conclusion & Next Steps

Robust image classification with 85–95% AI accuracy; ready for scale and refinement



**Project outcome:** Functional dual-approach classifier

Built hash-based and deep learning classifiers; supports batch processing and easy use.



**Performance:** Strong accuracy range

AI classifier achieves **85-95%** accuracy, suitable for practical deployments.



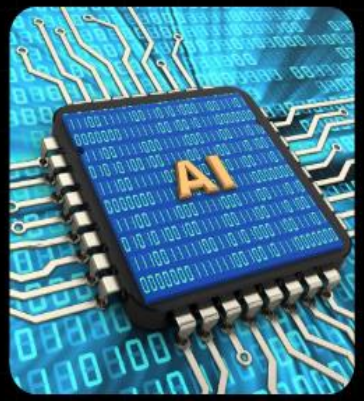
**Near-term focus:** Expand and refine

Add categories and refine models to boost accuracy further.



**Operational focus:** Optimize runtime and scale

Improve efficiency for real-time and large-scale deployments.




**Value:** Solid foundation for R&D

Provides a practical base for continued automated image categorization work.

# THANK YOU